

Final Report: SCUBA Diving in Square Lake September 2021 – September 2024

The goal of this project was to visualize and perform a statistical analysis of the SCUBA Dives that I have conducted in Square Lake. Each of those dives was logged in my dive log. Over the last 4 years, I have conducted 78 dives in various locations with 67 of them being in Square Lake. For each dive, I log the following information in my Physical and Spreadsheet Dive log: Dive Number, Date, Location, Dive Site, Time In, Bottom Time, Total Time, Max Depth, Visibility, EANx (Enriched Air Nitrox), Current, Waves, Water Temp, Suit (Thermal Protection), Tank, Weight, and Dive Notes. To consolidate and prioritize columns in the data frame, I only used the following information: Bottom Time, Max Depth, Temperature, Visibility, and Lead Weight Worn. For this project, I used the following technologies to create this project:

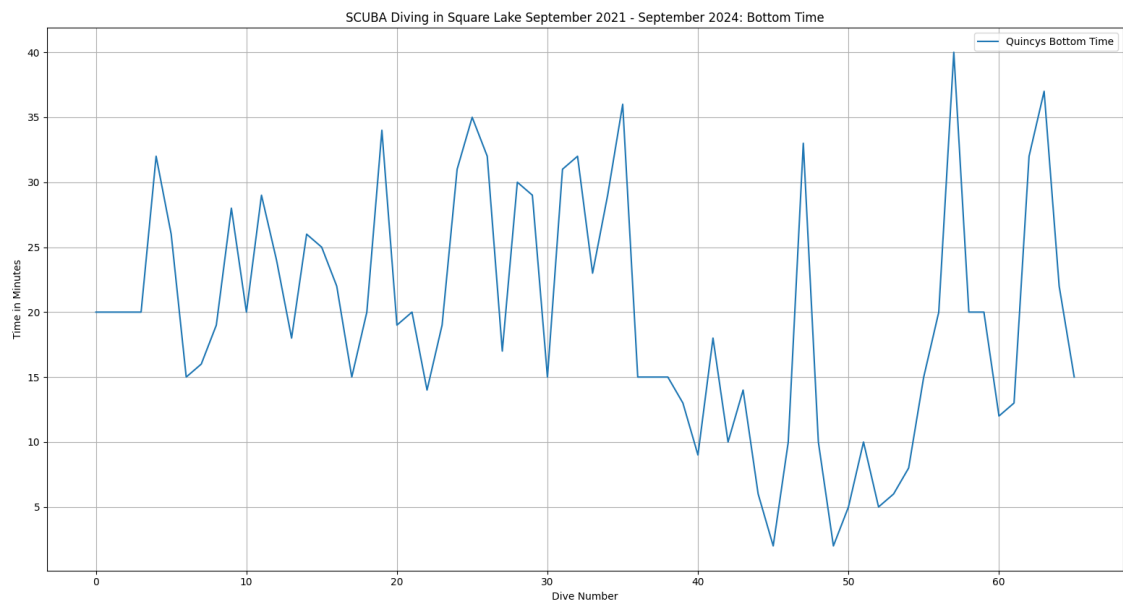
- Python3 (Matplotlib and Pandas)
- Visual Studio Code
- LibreOffice Calc

The reason I created this project was after completing my bachelor's in computer science from the University of Wisconsin–Stout, I started to build my portfolio. In addition, I started my freelance business: SCUBA Q Multimedia. My portfolio consists of the following areas: Motion capture, Software, Theatre Productions, Data Analytics, SCUBA, and Podcasts. One area of Computer Science I enjoyed learning about was data and statistical analysis using Python with Professor Yida Bao. Thus, I figured for a portfolio project I could combine two areas

I’m passionate about: SCUBA Diving and Data Analysis. This is the report of my findings from this project.

Data Visualization of Dives:

Bottom Time:



Mean: 19.893939393939394

Variance: 84.83473193473193

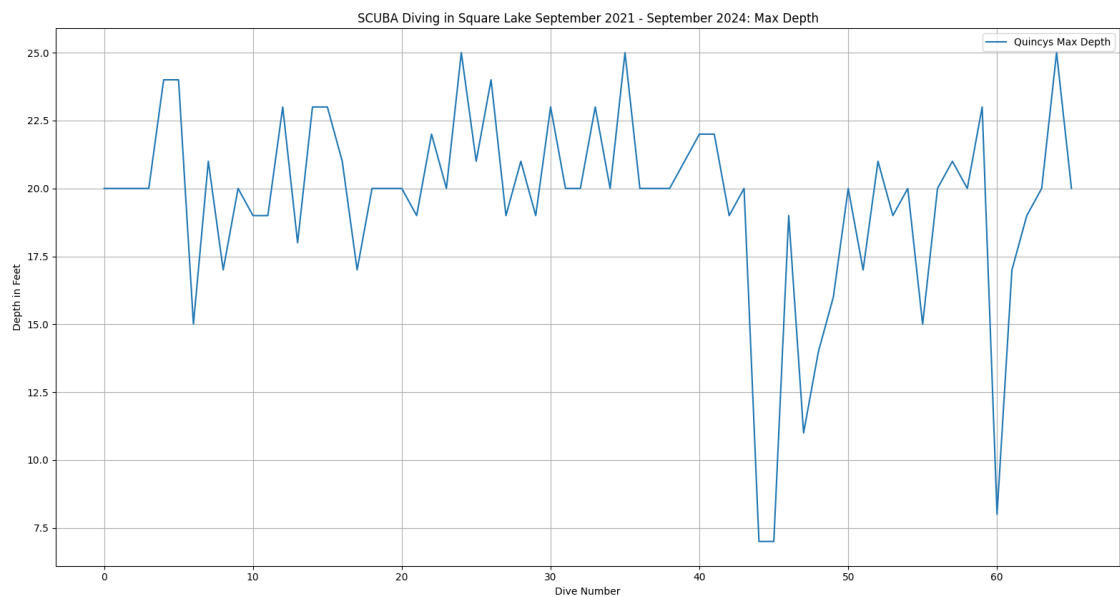
Median: 20.0

Mode: 0 20

Sum (Minutes): 1313

Sum (Hours): 21.883333333333333

Max Depth:



Mean: 19.515151515151516

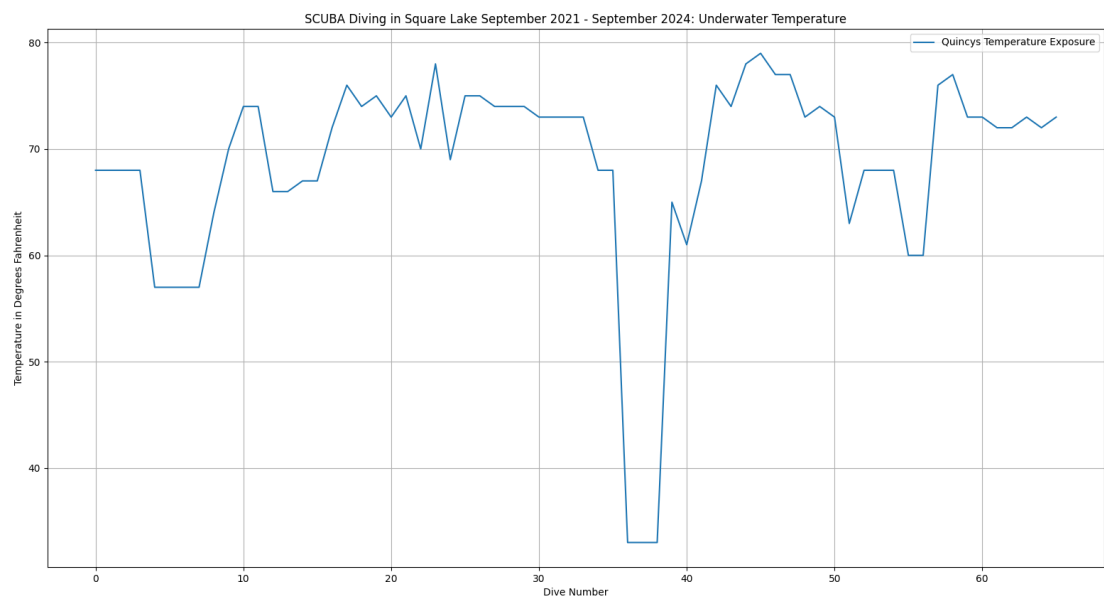
Variance: 13.761305361305359

Median: 20.0

Mode: 0 20

Sum (Feet): 1288

Temperature:



Mean: 68.65151515151516

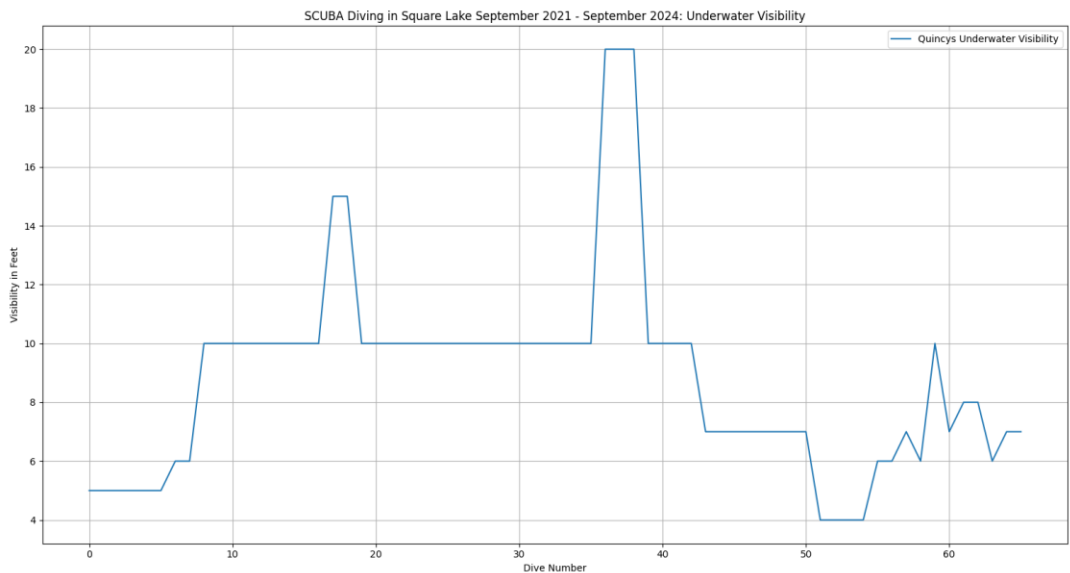
Variance: 91.64592074592073

Median: 72.0

Mode: 0 73

Sum (Degrees Fahrenheit): 4531

Visibility:



Mean: 8.818181818181818

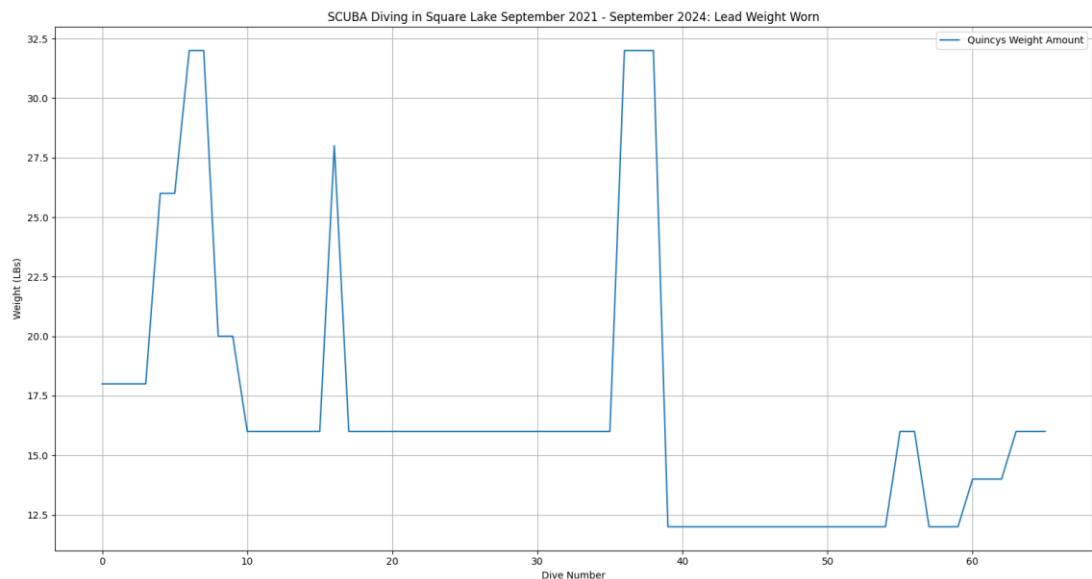
Variance: 11.751048951048954

Median: 10.0

Mode: 0 10

Sum (Feet): 582

Lead Weight Worn:



Mean: 16.696969696969695

Variance: 30.091375291375293

Median: 16.0

Mode: 0 16

Sum (LBs): 1102

Conclusion:

After using Matplotlib to graph the previously mentioned columns from my Square Lake Dive Log, the result has determined a relationship/trend between each column based on the dive number in the dive log.

The first example of this relationship/trend at work is when I did my first ever SCUBA dive. I was at 18 pounds (lbs) for lead weight worn with a 3mm wetsuit. Once I gained more diving experience and lost weight over the years, I managed to get the lead weight down to 12 pounds (lbs) with a 5mm wetsuit for my dives in 2024. Factors that play a huge role in the lead weight needed for a diver include Body Weight, Muscle to Fat Ratio, Thermal Protection, etc.

Another example of this previously discussed relationship/trend is depending on the season the dive is conducted and its dive number, this can result in a temperature increase or decrease. During dives 37, 38, and 39, I went for my PADI Ice Diving Certification which was conducted on January 21st. During this time, the temperature recorded in the lake was 33 degrees Fahrenheit. However, when I conducted dives 44 – 48 which were in August, the temperatures in the lake were the highest temp recorded being 79 degrees Fahrenheit.

In conclusion, there are more examples in addition to the previously discussed examples to support the relationship/trend between Dive Number and the columns. Therefore, Dive number will affect the other columns based on time of year/season, time of day, type of dive conducted, and goal of each dive.